Environment and Ecosystem

Lecture 1 comprises of the following

i. Environment Definition
ii. Earth-life support system.
iii. Ecosystem definition, the various components and types of ecosystem.

1.The Environment :

The word Environment originated from the French word **Environner (encircle or Surroundings**).

1.1: The Definition of Environment, as per Environment (Protection) Act, 1986

The sum of water, air, and land and the inter-relationships that exist among them and with the human beings, other living organisms and materials

From this **word etymology** we understand that environment means all that **surrounds** us.

So simply putting it together,

ENVIRONMENT is defined as the social, cultural and physical conditions that surround, affect and influence the survival, the growth, and the development of people, animals or plants.
.

2. Understanding the Terminologies

2.1 Environmental Science:-

Environmental science is the study of the environment, its biotic & abiotic component's & their relationship.

Wikipedia defines Environmental Sciences: as an interdisciplinary academic field that integrates Physics, Biology and Geography to the study of the environment, and the solution of environmental problems.

In simple words: **Environmental science** is an interdisciplinary study of how humans interact with the living and non-living parts of their environment.

2.2 Environmental Engineering:-

Environmental Engineering is the application of engineering principles to the protection & enhancement of the quality of the environment and to the enhancement and protection of public health & welfare.

2.3 Environmental Studies (or) Environmental education:-

Environmental studies is the process of educating the people for preserving the quality of the environment.

- **Scope *and Importance of* Environmental *Science***

3.1 Scope of Environmental Science:

1) To be aware and sensitive to the total environment & its related problems.

2) To motivate active participation in environmental protection & improvement.

3) To develop skills to identify & solve environmental related problems.

4) To know the necessity of conservation of natural resources.

5) To evaluate environmental programmes in terms of social, economic, ecological & aesthetic factors.

6) To promote the value & necessity of local, national & international co-operation in the prevention and solution for environmental problems.

7) To give a clear picture about the current potential of resources & environmental situations.

8) Environmental studies gives us an idea and understanding of the interdependent connection of nature and people.

3.2 Importance of Environmental Sciences

1) It has a direct relation to the quality of life we live.

2) People understand the need of development without destruction of the environment.

3).People gain knowledge of different types of environment & effects of different environmental hazards.

4) People are informed about their effective role in protecting the environment by demanding changes in laws and enforcement systems.

5) It develops a concern & respect for the environment.

- **Earth life support systems**

The earth system is itself an integrated system, but it can be sub-divided into four main components, sub-systems or spheres: the **geosphere, atmosphere, hydrosphere and biosphere**. These components are also systems in their own ways and they are tightly interconnected. Life is sustained by the flow of energy from the sun through the biosphere, the cycling of nutrients within the biosphere and gravity

The main components (called **spheres**) of the environment are:

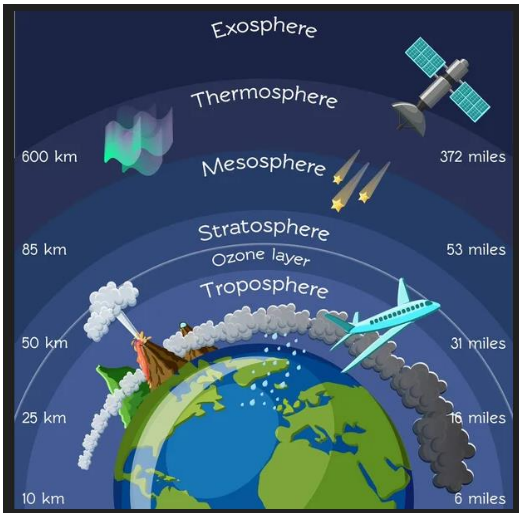
i. Atmosphere: The blanket of air that surrounds us.
ii. Hydrosphere: The various water bodies on the earth for eg the oceans, the rivers, lakes and ponds.
iii. Lithosphere talks of the various types of soil and rocks on the earth's surface.
iv. Biosphere: It contains all living organisms, their interactions with the environment and all that is capable of supporting life.

4.1.Atmosphere:

The blanket of air upto 1500 km surrounding the earth is known as atmosphere

4.1.1 Layers of the Atmosphere

Based on the distribution of temperature with height, our atmosphere is said to have the following layers.



4.1.2 Importance of the Atmosphere:

(i) Oxygen is very important for the living beings.
(ii) Carbon dioxide is very useful for the plants.
(iii) Dust particles present in the atmosphere create suitable conditions for the precipitation
(iv) The amount of water vapour in the atmosphere goes on changing and directly affects the plants and living beings. (v) Ozone protects all kinds of life on the earth from the harmful ultra violet rays of the sun.

4.2 Hydrosphere:is the discontinuous layer of water at or near the Earth's surface. It includes all liquid and frozen surface waters, and groundwater held in the soil

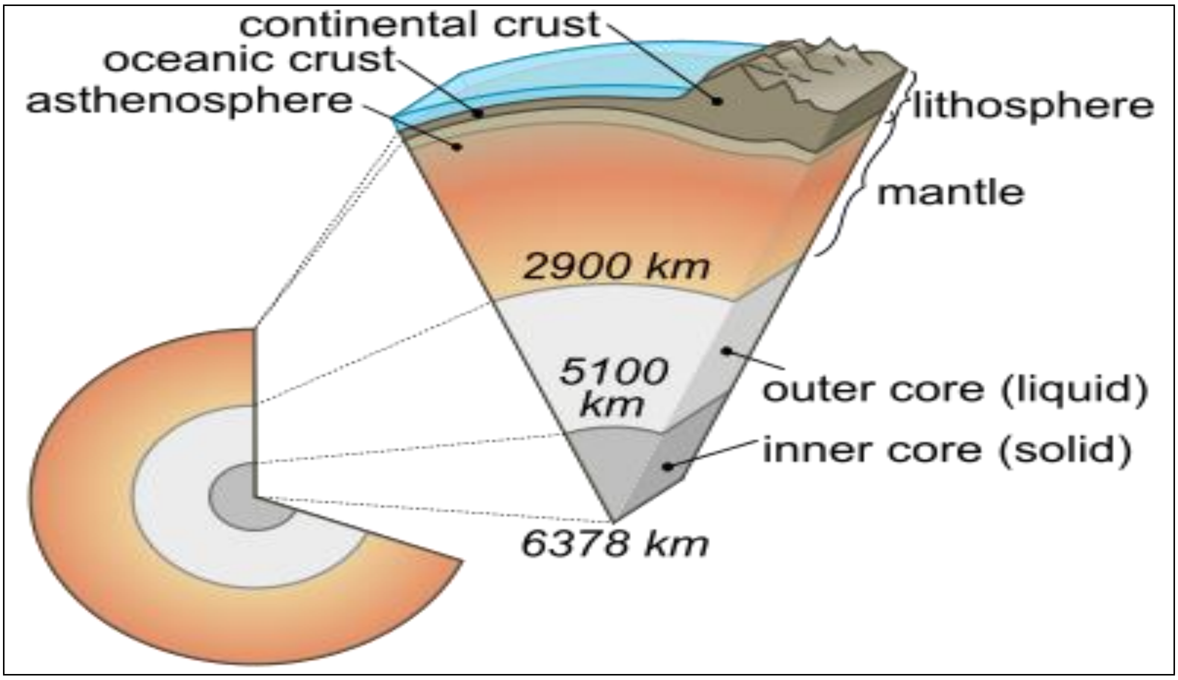
The existence of hydrosphere depends on an important phenomenon called the water cycle or the hydrological cycle.

4.2.1 Importance of the hydrosphere

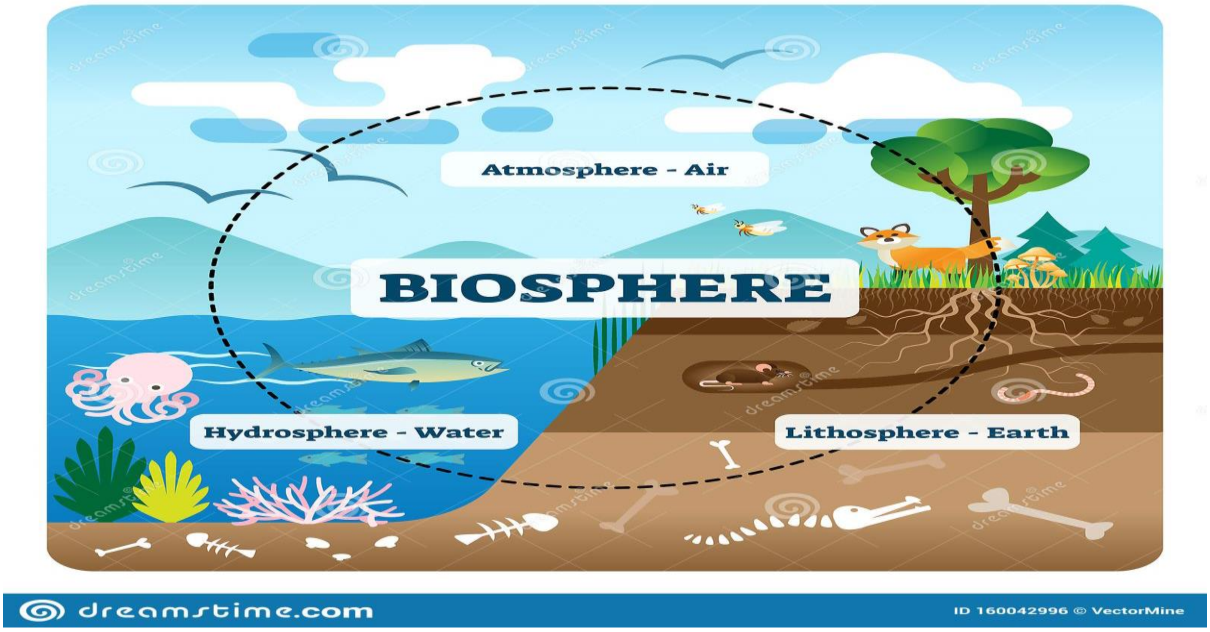
(i) One of the Basic Needs of Human
(ii) Part of a Living Cell
(iii) Habitat for Many Organisms
(iv) Regulates Temperature
(v) Atmosphere Existence

4.3Lithosphere: is the solid rock that covers the planet. This includes the crust, as well as the uppermost part of the mantle, which is the solid rock. The significance of the lithosphere is the activity of the tectonic plates.

The lithosphere



4.4The biosphere: is the zone where the lithosphere, the hydrosphere and the atmosphere interact with each other. This narrow sphere of the Earth supports life due to the presence of land, water and air. Therefore, the biosphere is important for living organisms as it supports life.



Source courtesy: dreamstime.com

These four components are main earth-life support system and constitute to make our earth A Living planet.

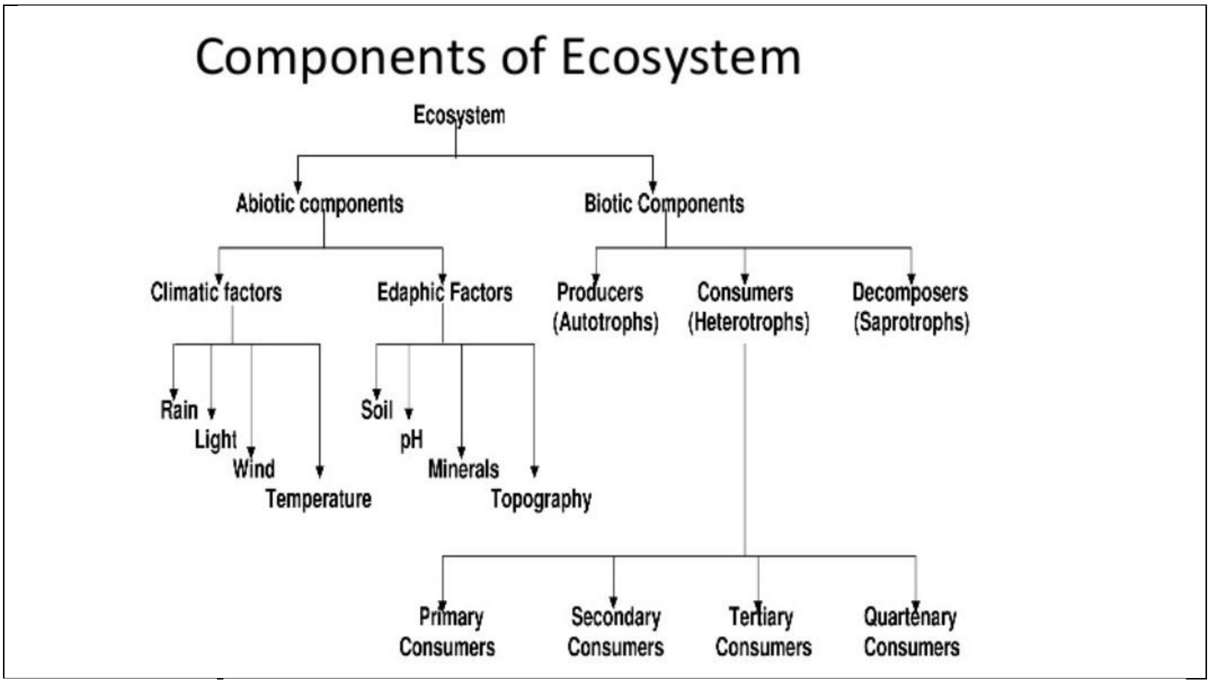
5. ECOSYSTEM

The term Ecosystem was first coined by **AG Tansley** in 1935. It is made up of 2 words: Eco meaning environment and system means a complex coordinated unit.

An **ecosystem** is defined as a natural unit that consists of the biotic components (living) and the non-living parts which interact with each other , probably allow exchange of materials to form a stable system. Eg. Pond ecosystem

Ecosystem is the basic functional unit of the organisms.

5.1 Structure and composition of an ecosystem:



Source courtesy: https://www.jagranjosh.com/general-knowledge/components-of-ecosystem

5.1.1 Ecosystems: Fundamental Characteristics

•**Structure**:

 Living (biotic)

 Nonliving (abiotic)

•**Process:**

 Energy flow

 Cycling of matter (chemicals)

• **Change**:

 Dynamic (not static)

 Succession, etc.

5.1.2 Components of an ecosystem ●

Ecosystem=biotic components + abiotic components

Abiotic Components: constitute the following

♦ Climatic factors: light, temperature, precipitation, wind, humidity ♦ Edaphic(soil) factors: soil pH, soil moisture, soil nutrients ♦ Topographic factors: aspect, altitude

Biotic Components: constitute the following
♦ Producers: green plants, algae
♦ Consumers: herbivores, carnivores, omnivores
♦ Decomposers: bacteria, fungi
Any ecosystem is made up of

Biotic Structure

- •**Producers(Autotrophs)** – Green plants which can synthesize their food themselves (Plants), chemoautotrophs

- .**Autotrophs** :• A groups of organisms that can use the energy in sunlight to convert water and carbon dioxide into Glucose (food) Autotrophs are also called Producers because they produce all of the food that heterotrophs use Without autotrophs, there would be no life on this planet **Examples**: Plants and Algae. **Photoautotrophs**(photosynthesis) **Chemoautotrophs**(chemical energy) Chemoautotrophs – Autotrophs that get their energy from inorganic substances, such as salt – Live deep down in the ocean where there is no sunlight – **Examples**: Bacteria and Deep Sea Worms

- **Consumers** – All organisms which get their organic food by feeding upon other organisms (Rabbit, man) **Heterotrophs** • Organisms that do not make their own food • Another term for heterotroph is consumer because they consume other organisms in order to live • Example: Rabbits, Deer, Mushrooms

- **Decomposers** – They derive their nutrition by breaking down the complex organic molecule to simpler organic compound (earthworms, ants).

5.2 Functions of an ecosystem:

- It regulates flow rates of biological energy.
- It regulates flow rates of nutrients, by controlling the production and comsumption of minerals and materials.
- It helps in biological regulation like nitrogen-fixing organism.

6.Types of Ecosystem

A natural ecosystem is a setup of animals and plants which functions as a unit and is capable of maintaining its identity. A natural ecosystem is totally dependent on solar energy. There are two main categories of ecosystems. They are:

6.1Terrestrial ecosystem – Ecosystems found on land e.g. forest, grasslands, deserts, tundra.

6.2 Aquatic ecosystem – Plants and animal communities that are found in water bodies. These can be further classified into two subgroups.

- Freshwater ecosystems, such as rivers, lakes and ponds.
- Marine ecosystems, such as oceans, estuaries.

All Ecosystems are either land-based (terrestrial) or water-based (aquatic)

6.1.1 FOREST ECOSYSTEM:

In the Indian continent, forests can be classified as coniferous and broadleaved forests. The type of the forests will depend upon abiotic factors such as soil, sunlight and soil nature in a particular region.

Depending upon the tree species: evergreen, deciduous, xerophytic and mangroves, forests classification can be attempted.

The structure and components of the forest ecosystem:

A. **Biotic Components**: The living components in a forest ecosystem are in the following order:

Producers: Different types of trees, shrubs and ground vegetation are the producers. Based on the climatic conditions, they are classified as: tropical, subtropical, temperate and alpine forests.

Consumers:

Primary: Herbivores such as ants, flies, spiders, dogs, beetles, elephants, deer, mongooses.

Secondary: Snakes, birds, foxes

Tertiary: Owl, peacock, lion, tiger.

Decomposers: Fungi and bacteria, essential to nature as they decompose the dead organisms of and release the essential nutrients for reuse.

B. **Abiotic components**: Soil, air, sunlight, inorganic and organic components and decaying organic matter.

6.1.2 GRASSLAND ECOSYSTEM

Grasslands are areas dominated by grasses. They occupy about 20% of the land on the earth surface. Grasslands occur in both in tropical and temperate regions where rainfall is not enough to support the growth of trees. The low rainfall prevents the growth of numerous trees and shrubs but is sufficient to support the growth of grass cover during the monsoon

Grasslands are found in areas having well-defined hot and dry, warm and rainy seasons.

Grasslands are one of the intermediate stages in ecological succession and cover a part of the land on all the altitudes and latitudes at which climatic and soil conditions (soil depth and quality) do not allow the growth of trees. The types of plants that grow here greatly depend on what the climate and soil are like.

Different Names of Grasslands

Grasslands are known by various names in different parts of the world.

The common ones are:

The Prairies of North America, The Steppes of Eurasia, The Savannas of Africa, The Pampas of South America,The Savanna of India and The Downs of Australia.

Tropical grasslands are commonly called Savannas. They occur in eastern Africa, South America, Australia and India. Savannas form a complex ecosystem with scattered medium-size trees in grasslands.

The structure and components of the grassland ecosystem:

Biotic Components

- **Producers** – In grassland, producers are mainly grasses; though, a few herbs & shrubs also contribute to the primary production of biomass.
- **Consumers** – In a grassland, consumers are of three main types:
    - **Primary Consumers** – The primary consumers are herbivores feeding directly on grasses. Herbivores such as grazing mammals (e.g., cows, sheep, deer, rabbit, buffaloes, etc), insects (e.g., Dysdercus, Coccinella), some termites and millipedes are the primary consumers.
    - **Secondary Consumers** – These are carnivores that feed on primary consumers (Herbivores). The animals like foxes, jackals, snakes, frogs, lizards, birds etc., are the carnivores feeding on the herbivores. These are the secondary consumers of the grassland ecosystem.
    - **Tertiary Consumers** – These include hawks etc. which feed on secondary consumers.
- **Decomposers** – These include bacteria of death and decay, moulds and fungi (e.g., Mucor, Penicillium, Aspergillus, Rhizopus, etc). These bring the minerals back to the soil to be available to the producers again.

Abiotic Components

- These include the nutrients present in the soil and the aerial environment.
- The elements required by plants are hydrogen, oxygen, nitrogen, phosphorus and sulphur.
- These are supplied by the soil and air in the form of CO_2, water, nitrates, phosphates and sulphates.
- In addition to these, some trace elements are also present in the soil.

Flora and Fauna of Grassland Ecosystem

- Grasses are the dominating plants, with scattered drought resistant thorny trees in the tropical grasslands.
- Badgers, fox, ass, zebra, antelope are found grazing on grasslands that support the dairy and leather industries.
- Grasslands also support the large population of rodents, reptiles and insects.

Functions of Grassland Ecosystem

- Energy flow through the food chain
- Nutrient cycling (biogeochemical cycles)
- Ecological succession or ecosystem development
- Homeostasis (or cybernetic) or feedback control mechanisms
- To increase the fertility of the soil and to regulate the productivity of the ecosystem.
- To reduce the leaching of minerals due to low rainfall.

Economic Importance of Grasslands

- Grasslands are the grazing areas of many rural communities.
- Farmers who keep cattle or goats, as well as shepherds who keep sheep, are highly dependent on grasslands.
- Domestic animals are grazed in the 'common' land of the village.
- Fodder is collected and stored to feed cattle when there is no grass left for them to graze in summer.
- The grass is also used to thatch houses and farm sheds.
- The thorny bushes and branches of the few trees that are seen in grasslands are used as a major source of fuelwood.
- Overgrazing by huge herds of domestic livestock has degraded many grasslands.
- Grasslands have diverse species of insects that pollinate crops.
- There are also predators of these insects such as small mammals like shrews, reptiles like lizards, birds of prey, and amphibia such as frogs and toads.
- All these carnivorous animals help to control insect pests in adjoining agricultural lands.

Classification of Grasslands

As climate plays an important role in the formation of grasslands, it is generally used as a basis to divide the world's grasslands into two broad categories: those that occur in the **temperate region** and those that occur in the **tropical regions.**

Tropical Grasslands

- These occur on either side of the equator and extend to the tropics.
- This vegetation grows in areas of moderate to a low amount of rainfall.
- The grass can grow very tall, about 3 to 4 metres in height.

- Savannah grasslands of Africa are of this type.
- Elephants, zebras, giraffes, deer, leopards are common in tropical grasslands

Temperate Grasslands

- These are found in the mid latitudinal zones and in the interior part of the continents.
- Usually, the grass here is short and nutritious.
- Wild buffaloes, bison, antelopes are common in the temperate region.

Grasslands in India

- In India, grasslands are found as village grazing grounds (Gauchar) and extensive low pastures of dry regions of the western part of the country and also in Alpine Himalayas.
- Perennial grasses are the dominant plant community.
- In the Himalayan mountains, there are high, cold Himalayan pastures.
- There are tracts of tall elephant grass in the low-lying **Terai belt** south of the Himalayan foothills.
- There are semi-arid grasslands in Western India, parts of Central India, and the Deccan Plateau.
- Grasslands support numerous herbivores, from minute insects to very large mammals.
- Rats, mice, rodents, deer, elephants, dogs, buffalo, tigers, lions, ferrets are some common mammals of grasslands.
- In northeast India, the one-horned rhinoceros is amongst the threatened animal of grassland in this region.

6.1.3 Desert Ecosystems: are found in regions where the annual rainfall is in the range of 250 to 500 mm and the rate of evaporation is very high. Occupy about 30% of the land area. They are characterized by extremely hot days and cold nights. The desert soils have very little organic matter and are rich in minerals. The desert plants have adapted to the dry conditions by having few or no leaves.

The structure and components of the desert ecosystem:

Biotic components:Producers: include xerophytic plants like cacti, shrubs, bushes, grasses, few trees, mosses and lichens.

Consumers: Primary Birds, camel, mouse.

Secondary: Lizards, snakes, birds.

Tertiary: Jungle cats, jackals, panthers

Decomposers: Some fungi and bacteria.

Functions of desert ecosystem

The dry condition of deserts helps **promote the formation and concentration of important minerals**. Gypsum, borates, nitrates, potassium and other salts build up in deserts when water carrying these minerals evaporates. Minimal vegetation has also made it easier to extract important minerals from desert regions.

6.2 Aquatic Ecosystems

Aquatic ecosystem is a water-based habitat. Many organisms rely on water for their livelihood and other life functions.

- The aquatic ecosystem is the basic functional unit facilitating the sustenance of aquatic organisms.
- The unique physicochemical features of this ecosystem allow the material transfer, carrying out significant chemical reactions, and other key functions needed for the survival of the life forms.
- Nekton, plankton, and benthos are some of the most prevalent aquatic creatures.
- Lakes, oceans, ponds, rivers, swamps, coral reefs, wetlands, and popular examples of freshwater aquatic ecosystems.
- While marine habitats include oceans, intertidal zones, reefs, and the seabed.

Types of Aquatic Ecosystems:

6.2.1 Freshwater ecosystems only cover about 1 percent of the earth's surface.

- Lakes, ponds, rivers and streams, marshes, swamps, bogs, and ephemeral pools are all . examples of freshwater.

- Freshwater ecosystems are divided into three types: lotic, lentic, wetlands, and swamps.

Lentic habitats are bodies of standing water such as lakes, ponds, pools, bogs, and other reservoirs. Flowing water bodies such as rivers and streams are represented by **lotic** ecosystems.

Lotic: Lotic ecosystems primarily refer to unidirectional, quickly flowing waterways such as rivers and streams.

- Several insect species, such as beetles, mayflies and stoneflies, as well as several fish species, such as trout, eel, and minnow, live in these settings.
- These ecosystems also include mammals such as beavers, river dolphins, and otters, in addition to aquatic species.

Lentic ecosystems:encompass all ecosystems with standing water.

- The principal examples of the Lentic Ecosystem are lakes and ponds.
- The term lentic is used to describe water that is stationary or relatively still.
- Algae, crabs, shrimps, amphibians like frogs and salamanders, rooted and floating-leaved plants and reptiles like alligators and other water snakes can all be found in these

6.22Marine Ecosystem: The marine environment covers the majority of the earth's surface area.

- Oceans, seas, the intertidal zone, reefs, the seabed, estuaries, hydrothermal vents, and rock pools make up two-thirds of the earth's surface.
- Aquatic animals cannot exist outside of water.
- Salt concentrations are higher in the marine habitat, making it difficult for freshwater creatures to survive.
- In addition, marine species are unable to survive in freshwater.
- Their bodies are designed to survive in salt water and will swell if placed in less salty water due to osmosis..
- They can be further classified as ocean ecosystems, estuaries, coral reefs, and coastal ecosystems.

6.2.3 Ocean Ecosystems:

- The Pacific, Indian, Arctic, and Atlantic Oceans are the five primary oceans on earth.
- The Pacific and Atlantic Oceans are the largest and deepest of these five oceans.
- More than five lakh aquatic species call these oceans home.
- Shellfish, sharks, tube worms, crabs, turtles, crustaceans, blue whales, reptiles, marine mammals, seagulls, plankton, corals, and other ocean plants are just a few of the organisms that live in these environments.
- **6. 2.4 Estuaries Ecosystems**
- Estuaries are critical forms of natural habitats which are typically formed where the sea and the rivers meet.
- The transition from land to sea happens in this region.

- As a result, the water here is more saline in comparison to freshwater ecosystems but more dilute than the marine ecosystems.
- Estuaries have more economic importance as they are capable of trapping plant nutrients and generating quality organic matter in comparison to all other land-based ecosystems.
- Estuaries today have also become hot spots for recreational activities and scientific studies.
- Some examples are tidal marshes, coastal bays, and river mouths.

6.2.5 Coral reefs

Coral reefs are underwater structures built from the skeletons of marine vertebrate, and are also called corals.These are found in most of the world's oceans.

- These corals form reefs called hermatypic or hard reefs as they give out hard calcium carbonate exoskeletons that protect their structure and support important life functions. Sea anemones are classic examples of hard coral reefs. The other species form soft reefs that are comparatively flexible organisms like plants and trees. Sea fans and sea whips are some of the most found varieties of soft reefs.
- The environmental conditions needed for the survival of coral reefs are warm, shallow, clear, and moving waters with ample sunlight.
- The Great Barrier Reef in Australia is the world's largest coral reef with a length of approximately 1500 miles.

6.2. A.Functions of Aquatic Ecosystem

- Allows nutrients to be recycled more easily.
- Aids in the purification of water
- Recharges thegroundwater table
- Provides a home for aquatic vegetation and fauna.
- Prevents flooding